



The impact of AI on the labour market

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Abstract

This study explores the impact of artificial intelligence (AI) on the labour market, focusing on changes in job roles, skill requirements, and human resource (HR) practices. Unlike previous surveys that primarily addressed technological aspects, this research systematically integrates technological, organisational, and institutional perspectives. The literature review (2020–2025) shows that AI adoption is closely associated with rising demand for technical and interdisciplinary skills, restructuring of work roles, and widening wage gaps between AI-skilled and non-AI-skilled workers. At the same time, many jobs are not eliminated but transformed, as AI frequently functions in augmentation rather than pure substitution mode. The results suggest that the future of work in the AI era will be determined less by technology itself than by how organisations, institutions, and policies govern its implementation.

Keywords Artificial Intelligence · Human resources · Labour market

1 Introduction

The changes driven by artificial intelligence (AI) and digitalisation are particularly pronounced in the IT sector, which not only develops these technologies but also applies them at a rapid pace. This leads to significant shifts in job tasks, organisational structures, and types of employment, making IT a key area to observe broader labour market transformations. Companies increasingly use AI to enhance operational efficiency, often resulting in fewer stable positions and a rise in temporary or outsourced work arrangements. A clear example of AI adoption in practice comes from Deutsche Telekom: starting in 2015, the company collaborated with

ALMATO to deploy around 400 software bots, or RPA (Robotic Process Automation) digital robots. By 2016, these bots were automating more than 40 processes across 30 different applications—including email management, SAP, and ticketing systems—handling roughly 320,000 business transactions each month. By 2018, Deutsche Telekom had expanded this system to over 1,500 “Frontend Assistants,” supporting 4 million monthly transactions and performing tasks such as checking connections, entering orders, generating invoices, and processing cancellations. Trade unions have responded to these developments by seeking to regulate AI, representing emerging worker groups, and negotiating how such technologies are implemented. Ultimately, the institutional framework and the extent of employee participation in decision-making remain crucial factors in achieving fair and inclusive digitalisation (Doellgast, 2023).

Since the turn of this millennium, AI has been concentrated mainly in the hands of large companies in the IT sector, but over time, it has begun to spread to other sectors, including services, research, business, and professional activities. A study by Damioli shows that this expansion was not only quantitative but also structural (Fig. 1). AI innovations have become more accessible to younger and smaller businesses, while at the same time contributing strongly positive to innovation activities outside the AI domain itself (Damioli et al., 2025).

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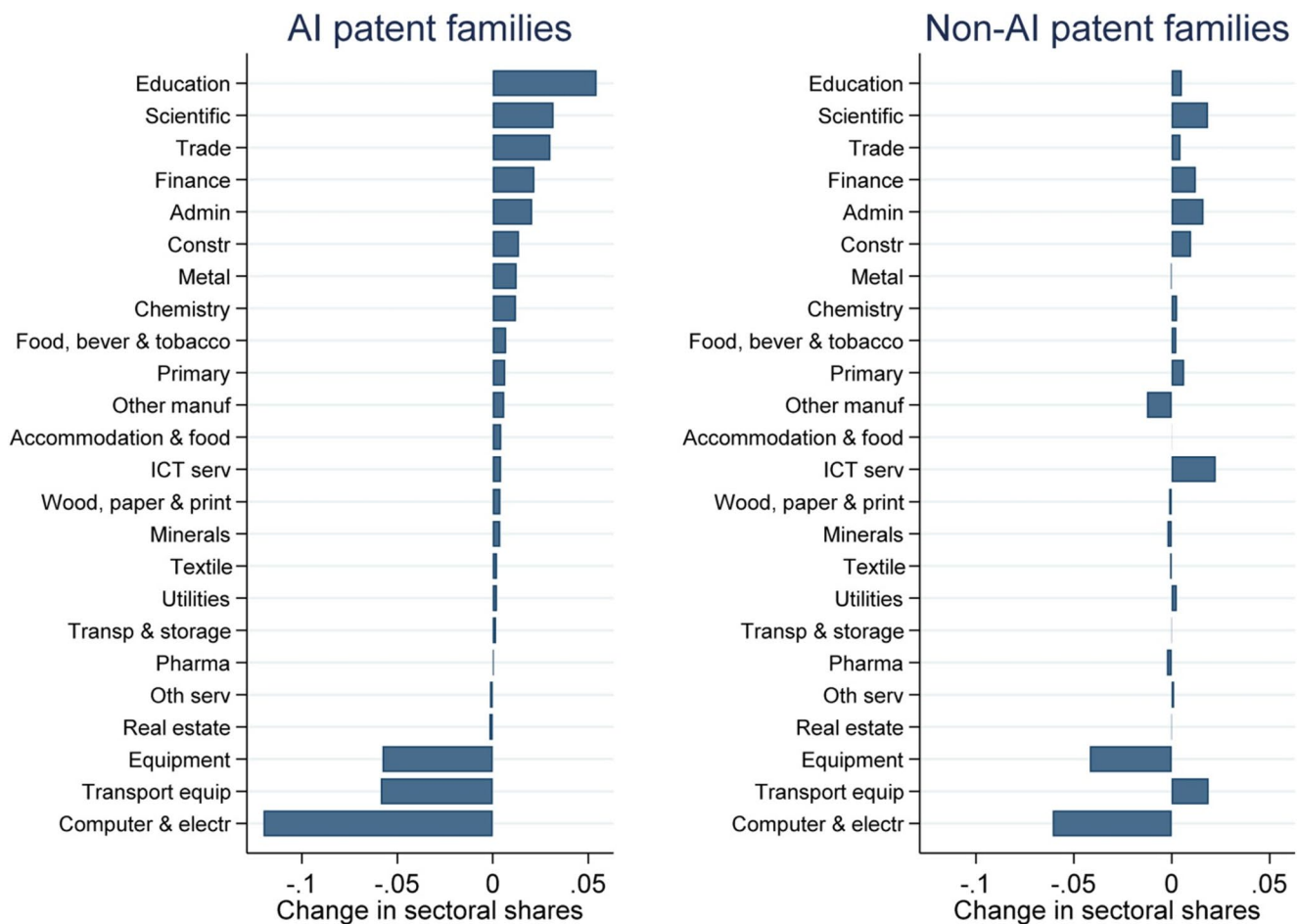


Fig. 1 Structural distribution of AI applicability across industries (Damioli et al., 2025)

Numbered aggregation step by step (from left to right): Bars are grouped into five aggregated industry blocks, numbered above each cluster: 1 – Core ICT manufacturing (computer and electronic products, electrical equipment, optical instruments); 2 – Other manufacturing (machinery, chemicals, metals, textiles, wood & paper, food, beverages & tobacco, transport equipment, other manufacturing); 3 – Core ICT services (ICT and scientific services); 4 – Other services (trade, transport & storage, finance & insurance, accommodation & food, real estate, other non-ICT services); 5 – Other industries (construction, utilities, primary industries).

To gain a more comprehensive understanding of these changes on a global scale, Damioli's study analysed a large micro-level database from PATSTAT, covering 23,915 applicants holding at least one AI-related patent between 2000 and 2016. Using text-mining techniques to identify patent families with AI-relevant keywords in titles or abstracts, the study revealed a rapid expansion of AI innovation: comparing the periods 2000–2005 and 2011–2016, the number of AI innovators grew by 338%, AI patent families

increased by 276%, non-AI patent families rose by 15%, and the ratio of AI to non-AI patents surged by 227%. Interestingly, however, the average number of AI patents per innovator decreased by 25%, suggesting broader participation but more focused patenting activity per entity. In terms of sectoral distribution, from 2000 to 2016 around 38% of AI patents originated from core ICT industries, with 27% in ICT manufacturing (computers, electronics, optical products, electrical equipment) and 11% in ICT services (publishing, audiovisual and broadcasting, telecommunications, IT, and other information services). Other important contributors included machinery manufacturing (13%), transport equipment (9%), and scientific R&D (10%). Notably, service industries played an increasingly prominent role, accounting for 39% of all AI patents, compared to just 15% of non-AI patents filed by the same innovators, highlighting the growing diffusion of AI beyond traditional IT sectors (Damioli et al., 2025).

These developments suggest that AI is becoming a key tool for generating and accelerating innovation across the economy, reinforcing its role as an “enabling technology”.

Thanks to its expanding application and ability to improve data-driven decision-making, AI is increasingly positioning itself as the foundation of a new technological regime, either within the existing IT paradigm or as a fundamental transformation of it.

The use of AI in predicting employee attrition illustrates a tangible, positive impact of AI on the labour market. A recent European study applied advanced machine learning methods to a unique cross-national survey covering 9,296 employees from 30 countries. The researchers compared both interpretable models, such as logistic regression, and state-of-the-art black-box models, including LightGBM (Light Gradient Boosting Machine), to predict employees' intention to leave their organizations within three months. Using robust evaluation procedures, including repeated 10-fold cross-validation and the Area Under the Precision-Recall Curve (AUC-PR) to account for imbalanced data, the study found that LightGBM and logistic regression were the top-performing models, achieving predictive performance more than twice the baseline (0.225). The analysis further identified the most influential factors driving turnover intention, including Sustainable Employability, Employership, Attendance Stability, Motivation, and Vitality. Causal analysis using structural causal models and partial dependence plots confirmed that improvements in these areas, particularly in employee motivation, are likely to reduce turnover. This approach enables organizations to anticipate attrition, plan staffing more effectively, and implement targeted interventions, highlighting how AI can support workforce stability while providing actionable insights for HR policies (Lazzari et al., 2022).

Some authors suggest that with the coming of advanced artificial intelligence, especially "General AI", there is not only a change in existing professions, but also the creation of entirely new job market offers and forms of work. With the ability to combine previously separate domains (e.g., healthcare and data science), new professions and fields are created. These changes cannot be explained simply by the automation of routine activities, but reflect a deeper transformation in innovation strategies, organisational learning, and the ability of companies to redefine the problem and solutions. This transformation of work reality emphasises interdisciplinary thinking, creativity, and the ability to navigate uncertain environments, redefining the concept of qualifications and competences in the labour market. Building on this idea, (Giacomoni, 2022) proposes a conceptual framework showing how AI can actively shape the innovation process itself. According to this framework, AI is not merely a tool for automating tasks, but a catalyst for *creating and transforming market offerings*, especially in complex sectors such as biotechnology and pharmaceuticals. By integrating AI into early-stage decision-making

and organizational learning, firms can explore previously uncharted solution spaces, hybridize knowledge from distinct domains, and adapt more flexibly to changing market conditions. This suggests that the arrival of General AI may accelerate the generation of novel products, services, and professional roles, supporting the notion that AI-driven innovation is closely intertwined with the evolution of work and the labor market.

2 Aim and methods

This research was created in response to the need to understand the dynamic changes that the increasing implementation of artificial intelligence is bringing to the field of work and human resource management. Given the rapid speed of technological development and the diverse approaches to AI adoption in different sectors and types of organizations, it is essential to create a comprehensive overview of current knowledge.

This study aims to identify key trends in employment, skills requirements, and HR practices through a thematic analysis of published research from the last five years, while assessing institutional and industry differences in approaches to digital transformation. This overview provides not only a basis for the research itself but also contributes to the formulation of strategies on how AI can be implemented equitably and sustainably. A comprehensive approach that connects the technological, organisational, institutional, and social dimensions will make it possible to identify the key factors that determine whether AI will promote inclusive work transformation or, moreover, reinforce existing inequalities.

This study uses a qualitative approach based on thematic analysis of secondary data. The starting point is a set of academic studies and empirical research published between 2020 and 2025 that focus on the impact of artificial intelligence (AI) on the labour market, the transformation of job roles, skills requirements, and changes in Human Resource (HR) management. Web of Science was used as the main scientific database, which provides access to peer-reviewed and high-quality scientific resources.

We searched the Web of Science (WoS) database for scientific papers focusing on AI, Artificial intelligence, Machine Learning, Natural Language Processing, HR, Human Resources, Labour/Labor Market, Digital Transformation and Economic issues. Although there are other databases, our priority was to filter out only those papers with the highest impact in the area under study (therefore articles published on conferences or by the publisher MDPI were purposely excluded from the results – low reputation (Beall list). For example, the initial search query generated

using parameter topic “Human Resources” AND “Artificial Intelligence” AND language restriction to “English” more than 1200 records. In this way, we gradually entered requests for all of the above keyword combinations. In total, we obtained more than 3,000 records, and it was necessary to filter out only those that were truly related to the research issue and not just those that had the given keyword in the title or abstract. For the analysis of scientific papers, we relied on a modified PRISMA 2020 standard adapted to our study (Fig. 2), as well as on procedures used by other authors (Imani et al., 2025), (Hassin et al., 2024), (Biswas et al., 2024).

Publications were further analyzed based on relevance, subject focus, and type of study. After a further round of selection, the sample was reduced to about only 100 articles, and 44 of these were identified as the most relevant for detailed analysis.

The initial search strategy yielded more than 3,000 publications related broadly to artificial intelligence (AI). However, a substantial share of these studies addressed AI from a purely technical, computational, or engineering perspective without explicit relevance to labour market outcomes, employment structures, or human resource management. Given the interdisciplinary nature of AI research, an additional thematic delimitation step was therefore required to systematically identify studies situated at the intersection of AI and labour market analysis.

To achieve this, we employed VOSviewer (Fig. 3), a bibliometric visualisation tool, to conduct a co-occurrence analysis of keywords based on titles, abstracts, and author keywords. This approach allowed us to map the conceptual structure of the AI-related literature and to distinguish clusters focusing on labour market, employment, skills, human

resources, and organisational change from those centred on algorithmic development or technical optimisation.

The co-occurrence network was constructed using VOSviewer based on keyword extraction from titles, abstracts, and author keywords. A minimum occurrence threshold of five was applied to reduce noise in the network. The full counting method was used for co-occurrence calculation, and association strength normalization was applied for link weighting. The clustering structure was generated using the default VOSviewer modularity-based clustering algorithm.

Specifically, keywords such as *labour market*, *employment*, *jobs*, *skills*, *human resources*, *wages*, and *workplace* were used as focal points in the co-occurrence network. Publications located within or closely connected to these thematic clusters were retained for further screening, while studies positioned exclusively within technical AI clusters were excluded. This bibliometric filtering step substantially reduced the dataset to a thematically coherent subset of publications with direct relevance to the research problem.

Following this visual and thematic delimitation, the remaining publications were subjected to manual screening based on predefined inclusion and exclusion criteria and assessed using a modified PRISMA 2020 framework. This two-step process—combining bibliometric mapping with qualitative screening—ensured that the final sample captured the most relevant scholarly contributions addressing the impact of AI on labour market dynamics rather than AI development in general.

As we can see, at first glance it may seem that the issue of AI’s impact on the labour market has been extensively studied and analysed. However, this is not the case; in fact, there are very few scholarly research papers that have addressed this issue. This research aims to understand the recent impact

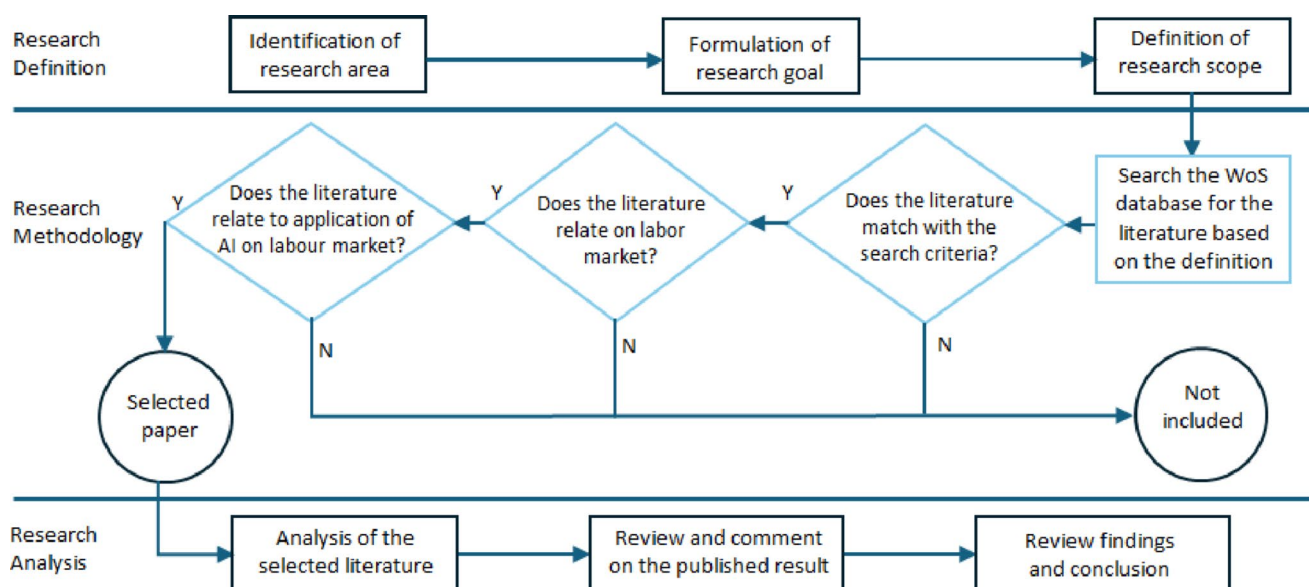


Fig. 2 Analysis of scientific papers with PRISMA standard

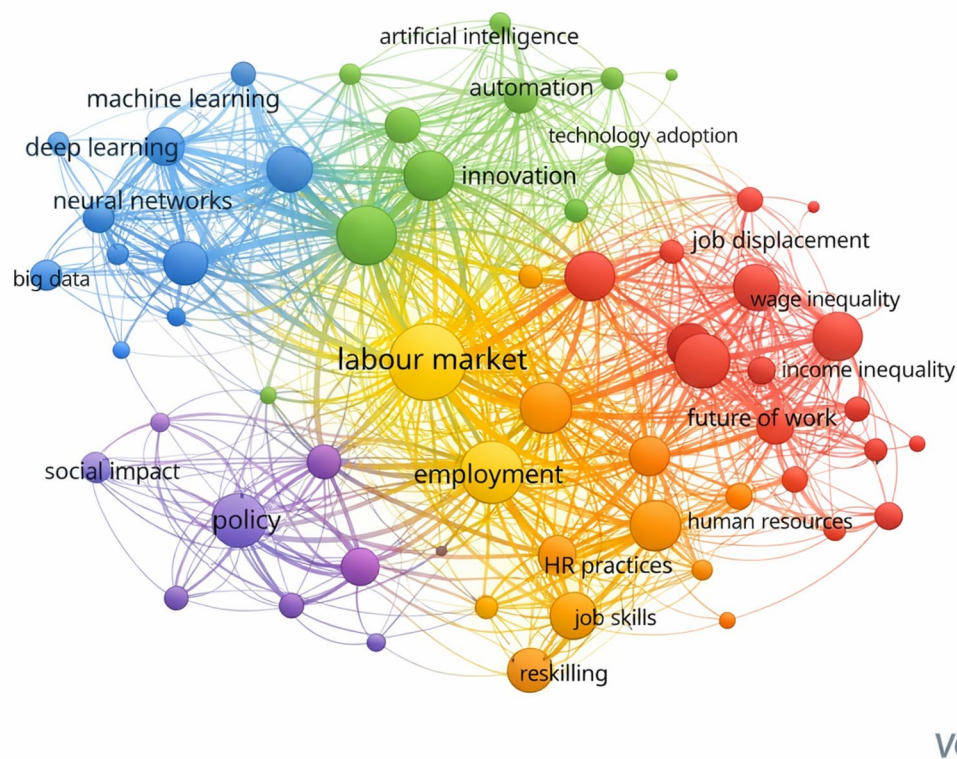


Fig. 3 Analysis of scientific papers with VOSviewer

of AI (analysed through analysis using WoS and PRISMA 2020 standard) on different aspects of the implementation of AI in the labour market and identify the primary challenges that limit its broader adoption.

The implementation of AI in HR management represents one of the strongest positive significant transformations of the current labour market. Despite the obvious benefits of AI, such as decision support, turnover prediction, or optimization of HR processes, its implementation raises serious questions about employment equity, disproportional distribution of benefits, gender and skill inequalities, and different approaches to employee retraining and participation. An important factor influencing the impact of AI is the institutional environment and the ability of organisations to involve employees in co-creating technological change.

Despite the abundance of scholarly literature and scientific papers, both in journals and at conferences, on the economic and technological aspects of AI, a comprehensive overview of how AI adoption affects job seekers, workers themselves, and interpersonal employment relationships, as well as skill dynamics across sectors, is lacking. Most existing studies rely on the rationale of technological progress, while overlooking the social, participatory, and institutional

dimensions of AI implementation. This study responds to this gap by examining how AI adoption works from an interdisciplinary perspective, with an emphasis on technological, organizational, and human aspects.

2.1 Research problem

How does the implementation of AI influence job structures, skill requirements, and equity in employment across sectors and organisations with different levels of technological maturity and institutional capacity?

The proposed research aims to explore how the impacts of implementing AI in HR vary between sectors and different types of organisations, with a focus on changes in employment, skills requirements and perceptions of fairness. The research will be based on a holistic approach that includes technological, organisational, institutional, and social aspects of AI implementation. This framework will identify the key factors that determine whether AI contributes to an inclusive transformation of the working environment or, on the other hand, leads to an increase in existing inequalities.

Based on the research problem, we set the research questions. Each of the proposed research questions addresses a critical aspect of how artificial intelligence (AI) transforms the labour market. Together, they provide a comprehensive lens for analysing the structural, organisational, and human-level implications of AI adoption.

RQ1: How are skill requirements evolving as a result of AI adoption in different sectors?

This question aims to identify the specific changes in the demand for skills driven by AI, such as the increasing importance of technical, data-analytical, and soft skills, and the obsolescence of certain routine competencies. Understanding these dynamics is essential for designing responsive training programmes, shaping educational curricula, and supporting workforce adaptability.

Contribution RQ1 *Offers insights for policymakers and educational institutions to anticipate future skill needs and align learning pathways accordingly.*

RQ2: To what extent does AI contribute to employment polarisation between high- and low-skilled workers?

This question explores whether AI accelerates the divergence in employment opportunities and earnings between highly skilled and low-skilled workers. It addresses the concern that AI may disproportionately benefit those already in advantaged labour market positions, thereby reinforcing existing inequalities.

Contribution RQ2 *Provides empirical evidence to inform equity-oriented labour market policies and contributes to the theoretical discourse on skill-biased technological change.*

RQ3: How does the institutional environment affect the implementation and outcomes of AI in the workplace?

This question investigates the role of institutions—such as trade unions, legal frameworks, and participatory decision-making structures—in mediating the effects of AI. The institutional context can either amplify or mitigate the risks of job displacement, inequality, and algorithmic control.

Contribution RQ3 *Highlights the importance of social governance and participatory practices, offering guidance for inclusive digital transformation strategies.*

RQ4: What strategies are companies using to address skill gaps and facilitate reskilling in the context of AI?

This question examines how organisations respond to AI-induced skill shortages—whether through investment in employee training, hiring strategies, or partnerships with

educational providers. It sheds light on organisational readiness and capacity to adapt to technological change.

Contribution RQ4 *Informs human resource development practices and fosters understanding of effective approaches to reskilling and workforce sustainability.*

RQ5: To what extent does AI implementation result in job displacement versus the creation of new roles?

This question assesses the balance between automation-driven job loss and the emergence of new positions requiring novel competencies. It aims to clarify the extent to which AI disrupts employment or facilitates occupational transformation and innovation.

Contribution RQ5 *Supports a more nuanced view of “technological unemployment” and provides data for forecasting future employment trends in the context of digital transformation.*

2.2 Analytical propositions guiding the thematic synthesis

Based on the research problem and research questions, this study formulates a set of analytical propositions that serve as organising devices for the thematic synthesis of the reviewed literature. These propositions are not hypotheses in the confirmatory sense, nor are they empirically tested within this study. Instead, they provide a structured lens through which convergence, divergence, and contextual variation across existing studies can be systematically examined. This approach is consistent with qualitative and integrative literature review methodologies, where propositions are used to guide interpretation rather than to validate causal claims.

AP1 Employee participation and job insecurity.

- **Rationale:** Employee participation and voice have long been recognised as central mechanisms shaping employee perceptions during periods of organisational and technological change trust (Doellgast, 2023). Recent contributions in the context of AI adoption highlight participation as a potential moderator of employee responses, including perceived job insecurity and acceptance of algorithmic management (Gómez Gandía et al., 2025). These strands of literature provide the conceptual grounding for treating employee participation as a key analytical dimension in the synthesis.
- **Analytical focus:** The synthesis examines how the reviewed studies conceptualise employee participation in AI-related decision-making and how they discuss its

relationship with perceived job insecurity and related employee outcomes across different organisational and institutional contexts.

- *Limitations:* Definitions and empirical operationalisations of employee participation and job insecurity vary substantially across studies, limiting direct comparability.

AP2 Asymmetric labour market effects of AI adoption.

- *Rationale:* The notion that technological change produces uneven labour market effects across occupations and skill groups has been extensively discussed in the economics of automation and task-based approaches to work (Zhang, 2023). In the AI-specific literature, the distinction between task substitution and task augmentation has become a dominant analytical framework for understanding employment effects (Johnson et al., 2022). This body of work motivates the inclusion of asymmetry as a central organizing proposition.
- *Analytical focus:* The review analyses how studies characterise asymmetric employment effects of AI, focusing on differences across skill levels, task types, and occupational groups, and how these patterns are interpreted within different empirical settings.
- *Limitations:* Empirical evidence is fragmented across sectors and methodologies, making it difficult to generalize findings beyond specific contexts.

AP3 Reskilling strategies and organisational adaptation.

- *Rationale:* Organisational adaptation to digital and AI-driven change is frequently framed around reskilling and continuous learning strategies in both HRM and organisational change research (Theben et al., 2025). Recent AI-focused studies emphasise training investments as a key mechanism through which firms seek to mitigate displacement risks and support workforce transformation e.g. (Duch-Brown et al., 2022). These contributions underpin the relevance of reskilling as an analytical dimension.
- *Analytical focus:* The synthesis explores how the literature discusses reskilling and training initiatives in relation to AI adoption, comparing internal training strategies with external labour market adjustments.
- *Limitations:* Many studies adopt a prescriptive stance and lack systematic evaluation of the effectiveness of reskilling initiatives over time.

AP4 Wage inequality and distributional outcomes.

- *Rationale:* Concerns about wage inequality and distributional consequences of technological change have a long tradition in labour economics e.g. (Cheng et al., 2024). In the context of AI, recent studies highlight skill-biased effects and differential returns to AI-related competencies as potential drivers of wage dispersion e.g. (Aleksieva et al., 2021), (Stephany & Teutloff, 2024). These debates justify the inclusion of wage inequality as a key analytical dimension of the review.
- *Analytical focus:* The review examines how studies analyse wage outcomes associated with AI adoption and how these outcomes are linked to skills, occupations, and institutional settings.
- *Limitations:* Differences in data sources, AI exposure measures, and definitions of AI-related skills limit the comparability of wage-related findings.

3 Literature review

In the literature review, we describe various approaches that are currently implemented as AI elements in the labour market. Artificial intelligence (AI) is a broad field that encompasses different computational approaches designed to perform tasks requiring human-like intelligence, such as learning, reasoning, or decision-making. Within AI, machine learning (ML) has become the most influential subfield, as it allows algorithms to improve their performance through exposure to data rather than explicit programming. Building on ML, a number of specialised application areas have emerged that are particularly relevant to the labour market and human resource management, such as Natural Language Processing (NLP) and Recommender Systems (RS). These methods can be very effective as an initial filter when submitting applications / CVs of applicants to larger companies, or for quickly determining the correctness of code for programmers.

3.1 Machine learning

For example, a study in China applied deep reinforcement learning (DRL) to predict mineralization potential in gold deposits, illustrating how ML can manage complex, high-dimensional datasets and make accurate predictions in uncertain environments. The dataset included 43 features per spatial unit, comprising 39 geochemical elements (such as Ag, Au, Pb, Zn) and 4 geological features (faults and stratigraphic layers), with data density of 4 samples/km². The data were split into training and testing areas at an 8:2 ratio to evaluate model performance. The DRL agent treated

each data point as a state and interacted with the environment to predict high or low mineralization potential. The model used ϵ -greedy policy and replay memory, with neural networks having three hidden layers (256, 64, 16 units). During training, the agent gradually learned to distinguish between known deposits and unlabeled areas. In the testing area, approximately 48.94% of the mineral potential areas identified by DRL with optimal parameters ($n=0.8$, $p=0.2$) contained around 80% of known gold deposits, while about 60% of these predicted areas covered 100% of known deposits, demonstrating strong predictive power. This example highlights that ML models, when properly designed, can leverage both labeled and unlabeled data to predict complex outcomes, even in dynamic or poorly understood systems. Similarly, in the labour market, predictive models trained on employee datasets (e.g., demographics, performance metrics, tenure) can forecast turnover probabilities, optimise staffing plans, and identify employees at risk of leaving, thereby improving workforce stability. The success metrics in these studies—such as the proportion of correctly identified deposits—are analogous to measures like prediction accuracy, recall, or precision in HR analytics, showing the tangible impact of AI-driven predictions. (Shi et al., 2023). ML thus provides the methodological foundation on which more specialised applications, such as NLP or RS, can be built. Machine learning thus provides the foundation for specialised applications such as NLP or recommender systems. In health data, models like random forests, SVMs, and neural networks are evaluated using metrics such as AUC, F1-score, and recall. High AUC values (closer to 1.0) indicate strong predictive ability, while 0.5 corresponds to random performance. These metrics allow ML to handle high-dimensional datasets like EHRs or genomic data and quantify predictive performance objectively. (Cho et al., 2024).

3.2 Natural Language Processing

Natural Language Processing (NLP) applies ML algorithms to textual and spoken data to understand, classify, or generate human language. Key HR applications include CV screening, employee feedback analysis, and sentiment monitoring. Modern NLP relies on pre-trained deep learning models, especially BERT and GPT variants, which achieve over 90% accuracy on benchmark tasks like sentiment analysis (IMDB: 50,000 reviews) and question answering (SQuAD-1.1: 100,000 questions; SQuAD-2.0: 150,000 questions including unanswerable ones). These models use large unlabeled corpora for pre-training and fine-tuning on target datasets, exploiting architectures such as transformers with self-attention, which outperform classical ML methods and RNNs. For instance, GPT-2 (1.5B

parameters) and GPT-3 (175B parameters) generate human-like text and enable zero- and few-shot learning. Despite their high performance, challenges remain in computational cost, interpretability, and reproducibility, as only 26% of recent papers report multiple fine-tuning runs, and 80% lack complete model selection procedures (Lauriola et al., 2022). Beyond traditional NLP tasks, these techniques have also been successfully applied to predictive optimization problems, where a study evaluated four vectorization and seven regression techniques to forecast application performance. Using NLP to analyze the input code, the framework predicts the optimal optimization sequence with a minimum performance drop of 0.5% and an average drop of 18% compared to the actual best sequence, demonstrating the ability of NLP-driven methods to identify highly accurate, task-specific solutions and highlighting their practical relevance for HR and other domain-specific applications (Ahmed et al., 2024).

3.3 Recommender Systems

Recommender Systems (RS) represent another major application area that builds directly on ML techniques. Unlike NLP, which focuses on language, RS use data on user preferences and behaviour to generate personalised recommendations. In the HR context, this includes suggesting suitable candidates for a vacancy, recommending training opportunities, or mapping career development paths. Experimental studies on movie recommendation datasets, such as MovieLens 100k and IMDB, demonstrate that combining PCA feature extraction, K-means clustering, and collaborative filtering yields highly accurate recommendations. For instance, clustering movies into six groups based on genre and incorporating user demographic attributes enabled the system to predict relevant items efficiently, achieving strong precision, recall, and F-measure scores while reducing runtime. These results highlight how RS can leverage ML techniques to generate targeted, high-quality recommendations, showing potential for analogous applications in HR decision-making (Yassine et al., 2021). Modern RS combine collaborative filtering with content-based and hybrid approaches, often supported by deep learning. Although RS are technically an application of ML rather than a distinct conceptual field, they illustrate how AI can be operationalised in human resource management to improve decision-making and employee development.

3.4 AI in relation to the labour market

This section synthesises the reviewed studies thematically to highlight shared patterns and divergences in the literature, rather than presenting isolated study-by-study

summaries. The findings are grouped into interconnected sub-themes reflecting key labour market dynamics associated with AI adoption. Empirical research using quantitative surveys with 420 responses (210 from business leaders, 210 from employees) shows statistically significant differences in the perception of automation across four areas: Business Processes, Technology and Tools, Skills and Professional Development, and Motivation and Engagement. For example, regarding workflow simplification (A1), business leaders rated it positively with a mean of 4.19 ($SD=0.394$), while employees rated it 4.03 ($SD=0.167$, $t(210)=5487$, $p<0.05$, $d=0.54$). Similarly, in Technology and Tools, business leaders saw higher benefits in tools efficiency (B1, $M=3.66$, $SD=0.476$) compared to employees ($M=3.20$, $SD=0.401$, $t(210)=10.647$, $p<0.05$, $d=1.04$). Differences were also notable in Skills and Professional Development, with work-life balance (C3) rated much lower by employees ($M=2.26$, $SD=0.441$) than by business leaders ($M=2.99$, $SD=0.138$, $t(210)=22.861$, $p<0.05$, $d=2.23$). Across Motivation and Engagement, large differences were observed in creativity (D1, $d=0.93$), career planning (D2, $d=0.82$), and skill obsolescence (D5, $d=0.8$). These results illustrate that while management emphasises efficiency and operational gains, employees are more concerned about job security, skill requirements, and personal development. Such divergent perspectives underline that AI adoption is not solely a technical process but also a socio-organisational one. Sustainable implementation, therefore requires participatory approaches, transparent communication, and targeted investment in competence development to align organisational objectives with employee needs and support long-term workforce sustainability (Gómez Gandía et al., 2025).

3.4.1 AI, human work, and organizational processes

Social skills are a key element of the added value of human work, as they enable effective collaboration, strengthen relationship networks, and contribute to productivity and salary growth. Evidence from the dataset on the joint use of AI and human resources (HR) demonstrates this clearly: even when AI and HR prices were equal, a minimum of 25% of daily labor hours was still required for AI to operate effectively. AI was able to reduce the remaining labor hours by approximately 50%, but the hours saved were not eliminated; instead, they were redeployed across other segments of work to maintain speed and quality. For example, for laborers with wage rates around \$124.50 per hour, 32 h of work per day were initially required, but after partial AI integration, 12 h were saved and reinvested into complementary tasks, while the total AI service bill matched the value of the saved labor hours. These findings confirm that AI primarily serves as a complement rather than a replacement for human

labor: it accelerates task completion and reduces repetitive effort, but cannot replicate the nuanced judgment, adaptability, and social skills that humans provide (Das, 2023). Integrating AI in HR improves efficiency and accuracy of decision-making through data-driven processes. AI tools such as machine learning, predictive analytics, and chatbots optimize workforce planning, talent management, performance reviews, and training. This transformation allows HR departments to focus on strategic activities while reducing the cost and time associated with routine administrative tasks (Benabou & Touhami, 2025). Chatbots are a modern HR tool that makes the recruitment process, communication, and administrative tasks easier and more efficient. They automate answering frequently asked questions, scheduling meetings, and collecting feedback, which evidence suggests reduces the workload of the HR department and enables consistent communication between candidates, employees, and recruiters. Implementing chatbots also transforms the way HR departments operate by allowing them to focus on strategic activities such as employee development and building a strong employee culture. Modern HR chatbots use artificial intelligence and machine learning to automate routine processes, increasing efficiency and reducing errors, although they still face the challenges of answering complex queries and the need for regular maintenance (Majumder & Mondal, 2021). While tools such as chatbots illustrate the operational integration of AI into HR processes, the broader economic implications of AI adoption depend strongly on how these technologies are priced and deployed within organisational decision-making frameworks.

Therefore although AI integration can significantly increase productivity and streamline decision-making processes, its ability to fully replace human labour is limited by several factors. These include the availability of data that may be incomplete or incorrect, technical limitations of current AI systems, and market conditions that affect the cost and effectiveness of automation. According to (Das, 2023) in linear pricing there is mass adoption of AI, which allows HR and AI to substitute for each other, while in the non-convex case, where AI is priced non-linearly, the innovator gains some monopoly share and there is complementarity between HR and AI. Overall industry demand is furthermore strongly influenced by the company with the highest preference for AI, showing how strategic decision-making and budget constraints shape substitution and complementarity between HR and AI technologies.

At an individual level, AI improves employee satisfaction, engagement, and overall performance by automating routine administrative tasks and freeing time for strategic activities. Empirical studies across 24 countries and 297 firms show that AI-enabled HRM applications provide personalized support, enhance learning, and reduce turnover,

while technological competence and firm-specific factors significantly influence AI adoption. From a business perspective, AI adoption leads to higher productivity, lower costs, and more efficient processes, boosting competitiveness. However, evidence also highlights risks such as increased uncertainty, stress, and the potential threat of human labour substitution.

Beyond organisational efficiency, the adoption of AI also affects employees directly at the individual level, influencing satisfaction, engagement, and perceived job security. Effective change management, socio-technical interventions, and AI-human configurations are therefore critical to balance efficiency gains with employee well-being and ensure positive organisational outcomes (Budhwar et al., 2022). Introducing technologies such as AI and robotics into HR brings greater efficiency and personalisation to processes such as recruitment, training and performance reviews. At the same time, it raises concerns about privacy, over-supervision and a decrease in employee motivation (Shahid et al., 2025). The implementation of AI in companies leads to higher labour cost inertia, suggesting that AI augments workers' capabilities rather than replaces them. As a result, companies rely more on skilled workers, which increases the costs associated with their adjustment and adaptation. To benefit from this change, companies should encourage continuous learning and adapt work processes so that both people and AI work effectively together. For example study analysed 28.463 firm-year observations of Chinese A-share listed nonfinancial firms from 2006 to 2020, using a textual measure of AI adoption based on annual reports. On average, firms mention 0.781 AI-related words per 1000 words, reflecting heterogeneous adoption across industries and time, with higher adoption in high-tech, IT, and capital-intensive sectors. Empirical results show that AI adoption significantly increases labor cost stickiness, indicating that AI tends to augment rather than replace employees, particularly in firms with higher-skilled workers, capital-intensive operations, and regions with older populations (H. Wang & Qiu, 2023). Digitisation and the expansion of AI are fundamentally changing the nature of work processes, especially in areas that have traditionally been seen as the domain of human creativity. Research indicates that instead of replacing human labour completely, it becomes reorganised. The automation of routine or technically specific tasks is changing the skill requirements of employees, reinforcing the importance of technological and data-analytical skills, while increasing the need for adaptive and interdisciplinary thinking. For example the study (Akmeraner-Kökat & Pellandini-Simányi, 2025), was based on a qualitative case analysis of a data-driven creative agency in Türkiye, drawing on 35 semi-structured in-depth interviews with copywriters, art directors, data analysts, and marketing specialists,

complemented by pilot interviews and industry media analysis. Using purposive and snowball sampling, the research examines how digitalisation and AI reshape creative advertising work practices. The study documents the widespread use of A/B/C testing, where alternative designs are exposed to samples of approximately 1,000 users, and selection is based on click-through and engagement rates. Metrics such as impressions, engagement and conversion rates increasingly govern creative decisions, shifting creatives' roles towards managing and responding to quantified feedback. Overall, the results show that AI does not replace human creativity but redistributes it, embedding human ideas into modular, data-driven systems that prioritise efficiency, scalability and measurable performance over artistic autonomy (Akmeraner-Kökat & Pellandini-Simányi, 2025).

While these organisational transformations reshape internal work processes, they also raise broader questions about employment dynamics, including job creation, displacement, and sectoral differences.

3.4.2 Employment effects: job creation, displacement, and sectoral differences

The deployment of AI not only improves productivity and the quality of work processes, but also raises questions about ethics, transparency, and potential job displacement (Aly, 2022). Econometric estimations confirm that digital transformation has a positive impact not only on economic development and labour productivity, but also on the overall employment rate. An interesting finding is that this positive effect is strongly reflected in female employment, suggesting that digitalisation may contribute to reducing gender inequality in the labour market. On the other hand, the effect of digital transformation on vulnerable employment is not found to be significant, suggesting that the reduction of vulnerable jobs depends more on other socio-economic and legal measures (Aly, 2022). The empirical analysis of 25 developing countries (2017) shows that digital transformation, measured by the Digital Evolution Index (DEI), is strongly and significantly associated with higher labor productivity and economic development. Specifically, a one-unit increase in DEI is linked to an increase of approximately USD 19,000 in labor productivity and a 1.3–1.45 rise in log per capita GNI, confirming productivity-driven growth effects. While descriptive evidence shows no clear unconditional relationship with employment, econometric results (FGLS) reveal a statistically significant positive impact of digital transformation on total employment, driven mainly by higher female employment. In contrast, no significant relationship is found between digital transformation and vulnerable employment, suggesting that technology alone does not improve job quality without supportive labor institutions. Overall, the results

highlight that digital transformation enhances efficiency and growth, but its social outcomes depend on complementary policies and change management (Aly, 2022).

In 2017, only 5.8% of U.S. companies used AI. However, when weighted by the number of employees, the adoption rate rises to 18.2% (see Fig. 4). This indicates that AI use is concentrated in large enterprises and fast-growing start-ups. The low average penetration rate suggests the existence of significant barriers and obstacles to AI adoption, which large enterprises have been able to overcome due to their size, thereby consolidating their competitive advantages. Analysis of start-ups reveals a strong association between AI use and process innovation, suggesting that innovative firms with high growth potential can drive the wider adoption of AI. Failure to tackle the current unequal tempo of AI adoption may result in increasing gaps in productivity, remuneration and career opportunities between tech-mature companies and small or less digitally mature businesses (McElheran et al., 2024).

Differences in the intensity of AI use across sectors suggest that the impact of this technology on the labour market will be very strong. As shown in Fig. 5, the highest intensity of AI use is observed in the information, healthcare and manufacturing sectors, where the proportion of companies with high intensity of AI use is greatest.

In contrast, sectors such as retail, transport, construction, or agriculture show overall lower adoption of these technologies. This difference is a result of the varying technological readiness, innovation capacity in each sector, and the specific character of the work they do. At the same time, it

means that the impacts of AI on employment, skills requirements, and inequalities will largely depend on the context of the specific sector.

The empirical results show that the impact of digitalisation on employees varies according to the type of technology. While exposure to robotics and software technologies often leads to worse working conditions, employees working with AI generally tend to have better job prospects. This finding supports the idea that AI, unlike previous waves of technological advancement, is not primarily focused on replacing manual labour but rather on supporting skilled tasks. (Serrano, 2025). The study also shows that the intense involvement of firms in global value chains generally has a negative effect on working conditions, but this relationship is not as significant as might be expected. Using microdata for over 9 million workers across 22 European countries, the estimates show that digital technologies and GVC participation independently affect working conditions, but their interaction is economically weak. Employment in highly digital-using sectors (MDIU) and in occupations exposed to software and robots is associated with significantly worse working conditions across all job-quality dimensions, while AI-exposed occupations exhibit better outcomes (e.g. positive coefficients around 0.004–0.010). Greater GVC intensity (measured by foreign value added in exports) is consistently linked to poorer working conditions, especially for skills, work intensity and prospects. Interaction terms between GVCs and digital exposure are often statistically significant but small in magnitude, indicating limited moderation effects. Overall, GVCs do not fundamentally alter

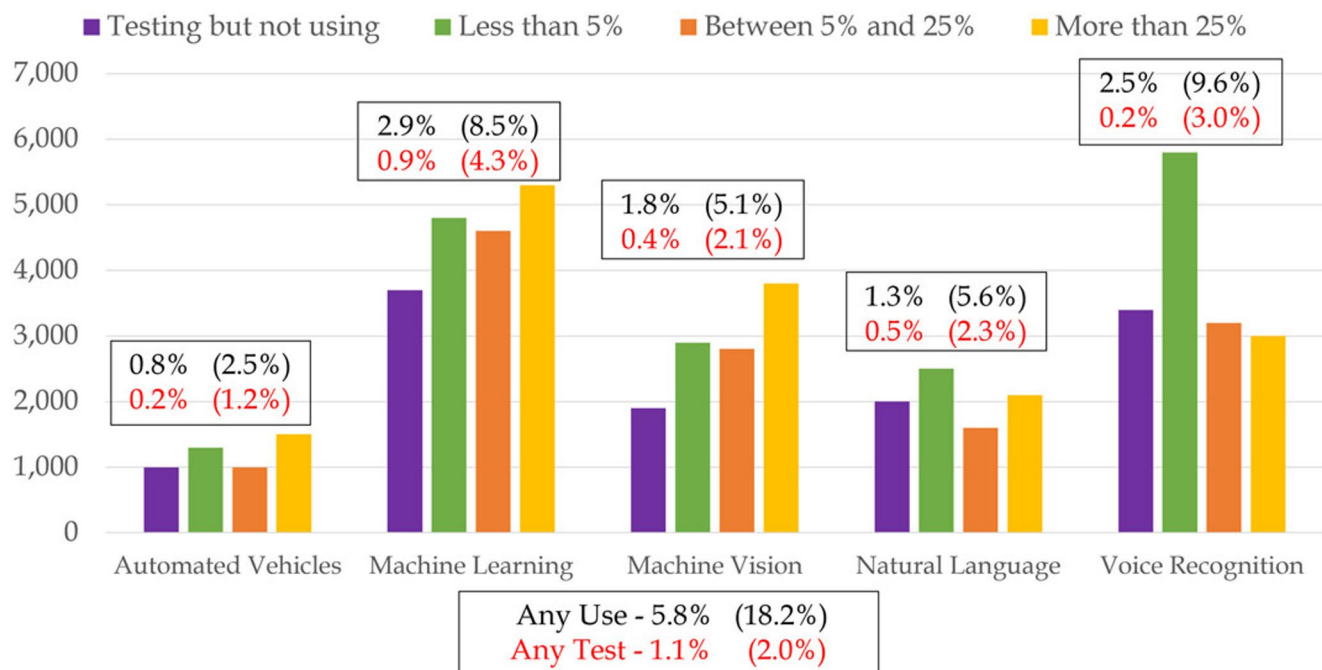


Fig. 4 AI adoption rate (McElheran et al., 2024)

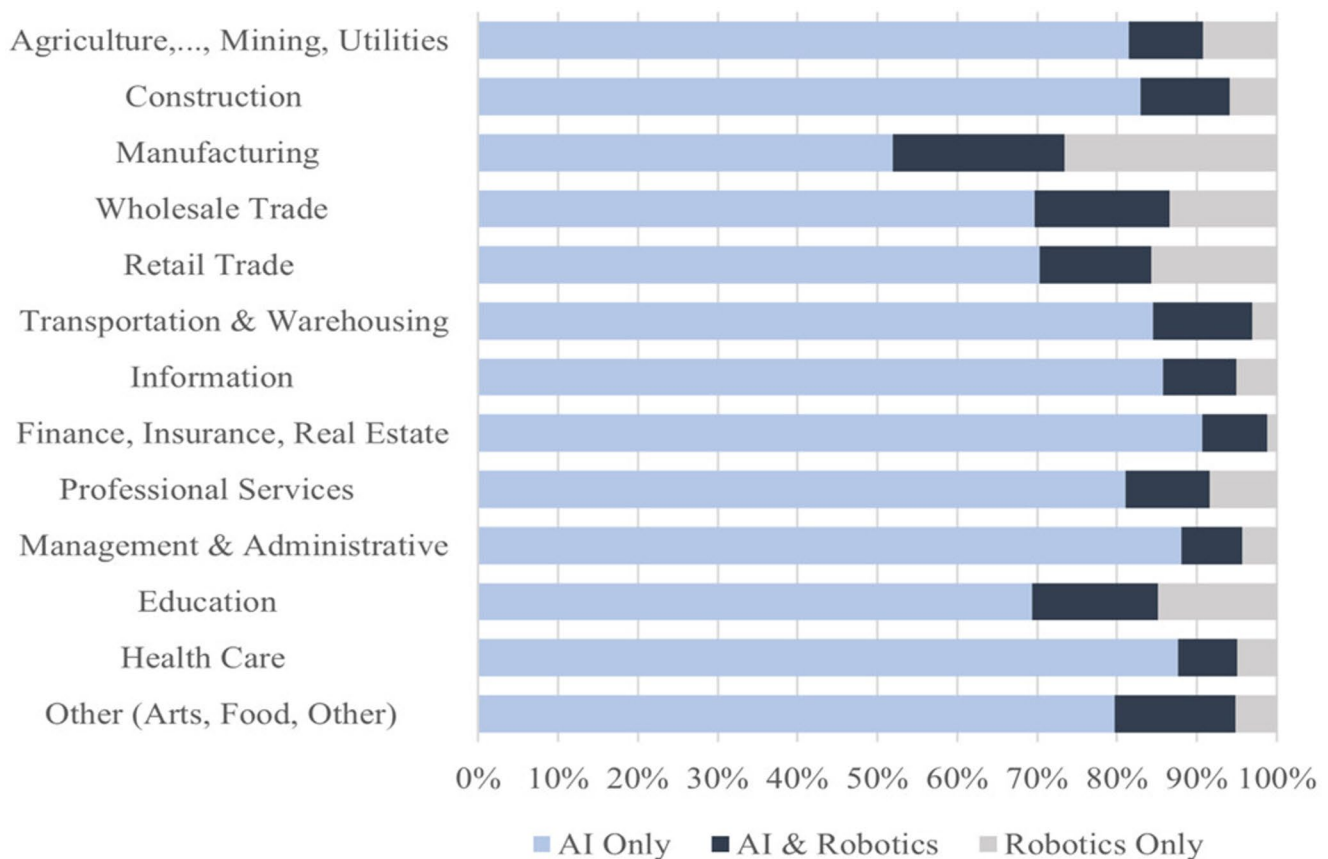


Fig. 5 Rate of AI usage across sectors (McElheran et al., 2024)

the technology–working-conditions nexus; instead, technology type (AI vs. software/robots) and sectoral position matter more than global production integration (Parteka et al., 2024).

Digital transformation brings fundamental changes to the labour market, but its impacts are not uniform. In technologically demanding sectors, gender inequalities persist, with women having limited representation and less accessible opportunities. While AI supports highly skilled tasks, it is unclear whether it will help eliminate this imbalance or rather deepen it. At the same time, it appears that companies involved in global value chains often exhibit worse working conditions, especially for women and younger workers, which can worsen their career prospects and salary opportunities (Parteka et al., 2024).

Empirical evidence supports this ambivalent impact of AI on firms and labour markets. Although AI-skilled workers account for only 0.3% of total employment, a 10% increase in AI investment raises firm growth by approximately 0.04%, mainly through productivity gains. At the same time, labour productivity positively affects growth (a 10% increase leads to a 0.09% rise in growth), while higher labour costs and labour share significantly reduce it. These results indicate that AI fosters growth when it complements

human skills and efficient labour market conditions, reinforcing the role of skilled, adaptable workers rather than fully substituting human labour (Tingbani et al., 2025).

AI currently has a complementary rather than a substitution effect on the research and development (R&D) labour market. Empirical analysis of 956 newspaper articles from 122 international sources (2020) shows that AI in R&D is used predominantly as a complement to human labour rather than a substitute. Approximately 90% of identified initiatives involve augmentation, while only about 10% focus on pure automation, indicating limited job displacement in R&D. The strongest concentration (43%) lies in the combination of augmentation and exploratory activities, confirming that AI mainly supports innovation, experimentation, and entry into new products and markets. This pattern is especially visible in healthcare, technology, government, and education, whereas manufacturing stands out for higher automation intensity. Overall, the findings suggest that AI is reshaping job roles and skill requirements, reinforcing the need for adaptability and human–AI collaboration rather than leading to large-scale employment losses (Johnson et al., 2022).

The degree of automation is influenced by a range of economic, political, and social factors, with the comparison

between the cost of human labour and investment in technology serving as a key determinant. Companies face strategic decisions between cost reduction and productivity enhancement. Automation can function not only as a substitute for human labour but also as a complement to it. In cases where demand for products and services increases, this complementarity may lead to rising salaries and the creation of new jobs (Das, 2023).

3.4.3 Skills transformation, reskilling, and education systems

Between 2010 and 2019, the rapid diffusion of AI in the U.S. labour market significantly increased demand for AI skills across most occupations. Job-posting data show a strong wage premium for AI skills: 16% when controlling for occupation–CBSA and industry effects, rising to almost 20% with firm fixed effects, and remaining at 11% and 5% after additional controls for occupation–CBSA and job titles. Despite this decline, the AI premium is still more than twice as high as that for other software skills. The highest premiums occur in management occupations, highlighting the value of strategic AI understanding beyond technical implementation. Firms with higher market value, intensive R&D investment, and larger cash reserves are more likely to demand AI-skilled workers and pay higher wages, while routine roles such as customer service face greater AI substitution risk (Alekseeva et al., 2021).

Attracting and retaining skilled employees is a key strategic challenge for SMEs. AI in recruitment enables the automation of a range of processes from sourcing and candidate assessment to interview scheduling with greater efficiency and lower costs than traditional processes. Furthermore, research suggests that AI can help reduce bias in the recruitment process, for example by eliminating gender-specific language in job adverts, which promotes more equal access to employment opportunities (Black & van Esch, 2020).

The field of HR is undergoing a fundamental transformation due to digital technologies and artificial intelligence. While the core functions of HR (recruitment, training, motivation, and monitoring) remain the same, the way they are performed is changing - recruitment is supported by AI tools, training is delivered through micro-learning, and employee engagement is enhanced by gamification. There is a growing focus on HR sustainability from an economic, social, and environmental perspective. Digitalisation is not just a question of technology. It also requires a cultural transformation of organisations where innovation plays a key role not only in efficiency but also in sustainability. With increased use of smart technologies, there is a need to focus on employment and skills development policies to match new labour market demands (Kraus et al., 2023).

Research demonstrates that the need for digital skills varies according to the stage of digitalisation of different sectors and their stage of technological maturity. While the demand for technical competences dominates in the early stages of digitalisation, domain knowledge, cross-disciplinary collaboration and decision-making skills become increasingly important as AI systems become more complex. Workers across the hierarchy of organisations are thus facing pressure to develop both basic IT skills and advanced analytical and soft skills, for example in evaluating AI tools (Theben et al., 2025).

This shift is also mirrored in the rapid growth of AI-related positions in statistics between 2010 and 2022. AI statistical job postings increased 31-fold, far outpacing non-AI statistical jobs (3.12-fold), with annual growth peaking at 35.7% before 2016 and rising again to 47.1% in 2021–2022. The sector now includes 932 distinct roles, led by Data Analysts (16.1%), Machine Learning Engineers (6%), and finance-related roles such as Risk Analysts (3.3%). Statistics increasingly clusters with Computer Science (25.8%), Mathematics (16.8%), and Economics (13.8%), highlighting interdisciplinary AI applications. Recruitment emphasizes hard skills—machine learning, Python, SQL, and predictive modeling—while traditional tools like SPSS and Excel decline. Certifications in Project Management (8.8%) and Financial Risk Management (8.0%) are increasingly valued. Most roles require only a bachelor's degree (66.9%), though actual AI employees often hold master's (42%) or Ph.D. (33%) degrees. Geographic and industry demand varies: Washington leads with 32.6% AI talent, and sectors such as retail (37%), information (30.8%), and manufacturing (28.1%) show the highest AI integration. Overall, AI is reshaping statistical roles, combining technical skills with strategic, interdisciplinary applications (Liu et al., 2024).

Analysis of the Chinese labor market from 2006 to 2020 shows a clear rise in educational attainment: the share of workers with junior high education or below declined, while technical secondary, senior high, junior college, and bachelor's degrees or higher steadily increased across primary, secondary, and tertiary industries. Wage data indicate polarization: most earn below 5000 RMB, though the share earning 5000–10,000 RMB and above 10,000 RMB gradually rises. Empirical analysis using LLC, IPS, and Breitung unit root tests, benchmark regression with fixed/random/mixed effects, and a dynamic panel GMM model confirms AI has a lagged effect on labor, enhancing productivity for both low- and high-skilled workers but widening income gaps. High-skilled workers adapt more easily to AI, benefiting from flexible hours and market participation, while low-skilled positions are more exposed to automation. This creates uneven labor market effects, with AI driving both upgrading

and polarization, reflecting rising education, skill-based income inequality, and the increasing value of digital and technical skills (Zhang, 2023).

The different rates of impact of AI on labour markets in different countries reveal that the key factor is not only the technological maturity or sectoral composition of the economy, but above all the specific content of work activities. Thus, the skills of workers and their suitability for automation play an important role. This is particularly important in developing and transition economies, where AI has the potential either to accelerate modernisation or, on the contrary, to deepen inequalities (Carbonero et al., 2023). The study shows that the labour market impacts of AI are conditioned by the specific structure of work activities, highlighting the need for locally tailored strategies. To assess the impact of AI on labor markets in Lao PDR and Viet Nam, was used Suitability for Machine Learning (SML) scores, which quantify the technical feasibility of automating 2,069 detailed work activities in the ONET database. Individual-level data on 44 work-related skills were obtained from the STEP survey, and semantic similarity matching using SBERT linked these skills to the ONET activities, producing individual SML scores for each worker. The mean SML (mSML) measures destructive digitalization, while the within-occupation standard deviation (sdmSML) captures transformative digitalization. In Lao PDR, where 40% of workers are employed in agriculture, the mean mSML was -0.619 and sdmSML was low, indicating low susceptibility to automation and limited potential for task reorganization. In urban Viet Nam, dominated by service and sales occupations, the mean mSML was -0.398 and sdmSML was high, reflecting higher exposure to automation and greater opportunities for workplace transformation. Women were slightly more affected by AI, though differences by gender were not statistically significant, while younger workers in Viet Nam faced higher automation risk than their counterparts in Lao PDR. These results highlight that the key factor in AI's labor market impact is not just technological maturity or sectoral composition, but the specific content of work activities. The skills of workers and their suitability for automation are crucial, especially in developing and transition economies, where AI can either accelerate modernization or deepen inequalities (Carbonero et al., 2023). Consequently, labor market outcomes are conditioned by the structure of work activities, emphasizing the importance of locally tailored strategies, including targeted support for vulnerable groups, development of adaptable skills, and effective retraining programs to ensure a fair transformation.

Investments in AI are having selective impacts in the Chinese SME market. While overall labour demand remains relatively stable, there is significant substitution of low-skilled workers in private, non-state and high-tech

companies, which illustrates the growing substitution effect of AI. This fact is further confirmed by study examining the impact of AI inputs on labor demand in 127 Chinese SMEs (635 observations) using a questionnaire survey validated against official registries. A two-way fixed effects model was applied, controlling for enterprise, year, industry, and city effects, alongside age, revenue, R&D, and wage levels. Endogeneity was addressed via lagged AI inputs as instrumental variables in 2SLS and SYS-GMM models, and robustness was checked through alternative AI proxies, sample trimming, and subgroup regressions. Benchmark results show that AI inputs do not significantly affect overall labor demand, but heterogeneity analysis reveals a significant negative impact in non-state-owned, private, high-tech enterprises, in the western region, new first-tier cities, and low labor-intensive or late AI-adopting firms. Mechanism analysis indicates a substitution effect on low-skilled labor, a complementary effect on high-skilled labor in certain sectors, and a creation effect for AI-related positions, explaining the insignificant aggregate impact. These findings suggest that AI reshapes labor structure rather than reducing total employment, highlighting the need for targeted reskilling, training, and policy support to mitigate technological unemployment (Xu et al., 2024).

Augmented and virtual reality (AR/VR) technologies are an effective tool to prepare workers for the changes brought about by the entry of AI into the job market. While VR is useful for new skills training and onboarding, AR helps to expand the competencies of experienced workers on the job through real-time instruction. These tools are applicable across industries, from manufacturing and logistics to customer service or management, and can play a crucial role in adapting lower-skilled workers to new jobs. However, to maximise their potential, it is essential that companies realise their benefits and systematically incorporate them into practice (Harborth & Kümpers, 2022). Research confirmed a strong correlation between employment levels and selected educational factors, identifying the employment rate of recent vocational education graduates as a key indicator. Individuals possessing digital and communication skills acquired through lifelong learning demonstrate a higher likelihood of successful labour market integration. The study used a mixed methods approach, combining quantitative analysis of CEDEFOP data for 20 European countries (2021) with qualitative design science experiments using UiPath to examine skills needed for smart technologies. Regression analysis showed that the employment rate is strongly influenced by vocational training graduates (IVETG), explaining 83% of the variation, while other indicators were not significant. The qualitative experiment highlighted the knowledge, skills, and attitudes (KSA) required for roles such as service manager, business analyst, software

developer, tester, and end users of software robots in industry. The findings emphasize that technical and transversal skills are crucial for young people to transition quickly from education to work. The study also highlights the necessity of aligning higher education curricula with the demands of the digital economy, while systematically fostering both technical and soft skills (Crețu et al., 2025). A key factor for meeting labour market demand is developed education competencies, the formation of which is significantly influenced by educational policies and curricula, as well as active cooperation with companies. This fact confirm also the study used PLS-SEM analysis to examine how the educational system (ESR) and business sector (BSR) influence educational competencies (EC) and labor market demands (LMD) in the AI era. Path coefficients showed that both ESR ($\beta=0.384, p<0.01$) and BSR ($\beta=0.277, p<0.01$) positively impact EC, which in turn directly affects labor market demands ($\beta=0.400, p<0.01$). Indirect effects confirmed that ESR and BSR influence LMD through EC ($\beta=0.154$ and 0.111 , respectively), highlighting the importance of integrated collaboration between education and industry. Preparedness for AI-based careers (PCBA) showed the strongest direct effect on labor market demands ($\beta=0.428, p<0.01$), emphasizing the need for targeted AI training. The findings suggest that curricular reforms, interdisciplinary learning, ethical AI education, and practical industry experience are critical for aligning competencies with labor market needs. The study underscores that a combined strategy of education-industry collaboration and AI-focused career preparation is essential for developing a workforce ready for an AI-driven economy (Badulescu et al., 2025).

The demand for AI specialists significantly exceeds the supply, resulting in higher compensation for AI-related projects. The labour supply elasticity ranges from 0.062 to 0.154, indicating that a 1% increase in compensation leads to a 6.2–15.4% rise in the supply of workers. The highest elasticity is observed in projects requiring expert knowledge, suggesting more intense competition among employers for highly qualified AI professionals. The findings also reveal that a change in platform regulations (specifically, the introduction of a requirement to disclose the expected level of experience) reduced labour supply elasticity by discouraging less qualified applicants from applying to more demanding projects (Duch-Brown et al., 2022). The adoption of AI technologies across sectors is leading to a significant shift in the skill requirements of the workforce, both at the level of technical, cognitive and soft skills (Theben et al., 2025). This confirms that the labour market changes brought about by the advent of AI cannot be addressed in isolation, but require a systemic approach based on collaboration between educational institutions, employers and policy makers. Although there is a broad consensus on the

need for continuous training and reskilling of the workforce in the context of AI adoption, corporate practice often relies on individual employee initiative or external recruitment of new employees with the required competencies. However, this approach contributes to a widening structural mismatch between the skills demanded and those available on the labour market. Moreover, the limited elasticity of labour supply in AI-intensive industries increases pressure on employers and educational institutions to strengthen their cooperation in formulating effective, tailored training strategies (Theben et al., 2025).

These findings also underline the importance of human capital development as a key complement to technological progress. Thus, strengthening innovation capacity and increasing salary share is not a result of the implementation of AI itself, but of its combination with investment in people, their skills and their ability to interact with new technologies.

While some studies warn of the possible negative consequences of AI on employment, empirical data from the European space shows a different picture. Research focusing on the EU over the period 2012–2022 shows that jobs with higher exposure to AI technologies show above-average employment growth. However, this growth is heavily concentrated in countries with advanced innovation systems called Innovation Leaders (e.g. Sweden, Netherlands) and Strong Innovators (e.g. Germany, France). Conversely, in countries labelled Moderate or Emerging Innovators, the positive impacts of AI on employment are not evident. These findings confirm that the impact of AI on the labour market is strongly conditioned by the institutional and innovation context of a country. A key factor is the ability of society (companies, institutions and individuals) to use new technologies effectively. AI is thus not a neutral technology, but its effects are asymmetric, both between countries and between sectors and professions (Guarascio & Reljic, 2025).

3.4.4 Wage effects and labour market inequality

These shifts in skill demand also translate into changes in labour market rewards, influencing wage structures and income distribution.

AI is fundamentally changing the value of job skills and thus the shape of the labour market. Skills are not valued by the market in isolation, but their value depends on how well they complement other skills. Workers with AI competencies therefore earn a significant salary bonus (on average 21% higher than other skills), reflecting the growing demand for these skills and their widespread application across industries (Stephany & Teutloff, 2024). Besides concerns about worker substitution and labour market polarisation, some studies provide evidence of the positive impacts

of AI on labour income. An analysis 25,156 of Chinese firms (2010–2022) shows that AI adoption can lead to an increase in the salary share of company income, particularly through increasing innovation capacity and technological advances. The findings show a robust positive relationship: AI adoption increases labor compensation rather than displacing workers, supporting the complementarity between AI and human capital. Two key mechanisms are identified: AI stimulates innovation capacity, increasing R&D intensity, and promotes technological upgrading, measured via patent output, both of which enhance labor income share. Heterogeneity analysis reveals stronger effects in non-SOEs and non-heavy-polluting industries, highlighting the role of organizational flexibility and industry characteristics. Robustness tests, including 2SLS, Heckman corrections, and alternative variable measures, confirm the stability of these results. Overall, the study suggests that when coupled with investments in innovation and human capital, AI can support inclusive growth and higher labor compensation, with important implications for corporate strategy and policy design (C. Wang & Jiao, 2025).

This fact is also confirmed by other study that shows AI technologies increase salary compensation for higher skills, widening the gap between workers with different skill levels. Middle-skilled workers, whose tasks are often easy to automate, are most at risk. While inequalities may stabilise in the long term due to the rising productivity of even lower-skilled workers, current trends show that without supportive measures AI can further increase structural inequalities in the labour market (Cheng et al., 2024).

AI is changing industrial processes by automating tasks previously performed by humans through structural, functional, and behavioural simulations. These simulations replicate human thought, computing, and behaviour, enabling the creation of “intelligent machines”. However, most current technologies fall into the category of weak AI, i.e. systems designed for specific tasks (e.g. image or voice recognition) that do not have a human level of general understanding or independent thought. In contrast, strong AI refers to a hypothetical form of intelligence that could think, learn, make decisions, and solve new problems independently without human intervention similar to or better than humans. But the development of strong AI faces several ethical and technological obstacles. From the perspective of economic theories such as Marx’s conception of capital, AI has a dual impact on the labour market. It replaces some jobs (substitution effect), but at the same time, it creates new job opportunities, especially in technological and high-skilled sectors (Zhang, 2023).

3.4.5 Institutional, policy, and socio-technical perspectives

Findings indicate that the development of artificial intelligence (AI) technologies is not inherently efficient and may be constrained by coordination failures between firms and households. This can result in the economy being held in equilibrium with low AI penetration, which limits the potential benefits of these technologies. Rather than imposing a tax on AI, which could exacerbate this situation, (Ünveren et al., 2023) propose the use of targeted subsidies and an active governmental role as a coordinator to promote greater AI adoption. However, such policy interventions should be tailored to the specific stage of AI development in each country. For example, redistributive measures may have opposing effects on AI investment depending on whether a country is at a low or high level of technological diffusion.

Although most studies of AI’s impact on the labour market rely on technological determinism and quantitative projections, alternative approaches, such as social shaping theories of technology, emphasise that technologies are not neutral, and their impacts are not inevitable. The development and implementation of AI are the result of social, political, and economic decisions, not autonomous technological developments. This perspective challenges the dominant narrative of the inevitable replacement of human labour by automation and instead highlights the need to understand AI as part of broader power structures and institutional conditions. Predictions of massive unemployment often overlook the human agenda, the impact of work collectives, government policies, and the real complexities of implementing technology in specific organizational contexts (Howcroft & Taylor, 2023).

AI systems help to reduce costs, manage risk, and increase efficiency, which puts pressure to transform work roles and places new demands on human capital. Simultaneously, it is becoming clear that customers and organisations require contextual understanding and expertise that cannot be fully replaced by automated solutions. This highlights the importance of maintaining the human-in-the-loop principle in decision-making processes. To mitigate negative impacts and maximise benefits, educational institutions and employers should foster skills such as adaptability, agility and lifelong learning (Ashta & Herrmann, 2021).

Although AI is unlikely to lead to widespread technological unemployment, it is fundamentally changing the structure of the labour market and the way work is done. In response to these changes, policy proposals such as a basic unconditional income, systemic retraining or a tax on automation are emerging. There is also a critical reflection on the concept of technological determinism, which assumes that the development of technology takes place outside of social control. The main risks associated with algorithmic

labour management include the weakening of workers' autonomy, the loss of meaningfulness of work, the creation of responsibility gaps or new forms of hidden labour and digital exploitation. It therefore highlights the need for a proactive and value-based approach to the development and implementation of AI that considers not only economic efficiency but also social justice, the protection of labour rights and environmental sustainability (de Santoni, 2024).

There is a need to stimulate investment in AI and digital infrastructure, support the creation and development of digital industry clusters, strengthen social protection systems for vulnerable groups of workers and modernise education to support lifelong learning and the development of skills relevant to the demands of the AI economy. These measures are key to fostering inclusive economic growth, reducing structural unemployment and increasing the resilience of the workforce to the impacts of automation (Shen, 2024). Investments in AI have a positive effect on employment in industry, with the job creation effect outweighing the labour substitution effect. AI increases productivity and supports the expansion of production capacity, leading to the creation of new, more complex and skilled positions. This trend is particularly reinforced where automation is not yet fully developed, replacing mainly monotonous and physically demanding jobs (Shen, 2024).

4 Discussion

This study set out to explore how the adoption of artificial intelligence (AI) is reshaping the labour market, with a focus on evolving skill demands, employment structures, and institutional influences. The discussion integrates findings from recent literature (2020–2025) to assess each research question and analytical propositions.

AI and Evolving Skill Demands (RQ1, AP2, AP4).

A recurring theme across multiple studies is that AI implementation significantly transforms the structure of skill requirements, reinforcing the need for technical and interdisciplinary competencies. Liu et al. (Liu et al., 2024) show a clear shift in employer preferences toward practical AI-related “hard skills” such as Python, SQL, and machine learning, with traditional statistical competencies losing prominence. Similarly, Stephany et al. (Stephany & Teutloff, 2024) demonstrate that AI-related skills yield a 21% salary premium, confirming a clear market signal favouring these competencies (converges AP4). When interpreted through the lens of AP2, the reviewed literature reveals a recurring pattern of asymmetric employment effects. While several studies point to displacement risks in routine occupations, others emphasise task restructuring and augmentation, indicating that outcomes are highly context-dependent.

This is supported by Zhang (Zhang, 2023), who finds that the technological and organisational barriers to implementing pervasive AI are much lower in high-skill occupations, contributing to labour market polarisation. However, some findings moderate this view. For example, Johnson et al. (Johnson et al., 2022) report that AI is more often used to augment rather than replace existing work functions in research and development, supporting a more nuanced interpretation of AP2: while routine jobs may be displaced, many roles are restructured rather than eliminated.

Employment Polarisation and Inequality (RQ2, AP2, AP4).

The impact of AI on employment polarisation is confirmed by several sources. Alekseeva et al. (Alekseeva et al., 2021) found that AI-skilled employees receive significantly higher compensation, not only for direct AI tasks but also in adjacent roles. Cheng et al. (Cheng et al., 2024) show that middle-skilled workers—whose tasks are easiest to automate—are particularly vulnerable to job loss, suggesting that inequality is growing both vertically (by skill level) and horizontally (within sectors). Moreover, McElheran et al. (McElheran et al., 2024) report that only 5.8% of U.S. companies used AI in 2017, but the employee-weighted usage was 18.2%, reflecting a high concentration of AI adoption in large firms. This supports the claim that AI reinforces structural asymmetries, especially between digitally mature and lagging firms or regions, lending further converges to AP4. Conversely, Guarascio & Reljic (Guarascio & Reljic, 2025) offer evidence that in some European regions, employment growth is positively correlated with AI exposure, but this effect is concentrated in “Innovation Leader” countries. This asymmetry supports the idea that institutional and innovation capacities moderate the impact of AI, connecting directly with RQ3 and AP3. These patterns mirror the heterogeneous adoption trajectories described in Sect. 3.4, where sectoral and firm-level differences strongly condition labour market outcomes.

Institutional context in shaping the social outcomes of AI (RQ3, AP1, AP3).

Doellgast (Doellgast, 2023) highlights how employee involvement in decision-making and the presence of strong unions contribute to more equitable outcomes in the digital economy. This aligns strongly with AP1, suggesting that participatory governance leads to reduced job insecurity and higher satisfaction. Similarly, Gómez Gandía et al. (Gómez Gandía et al., 2025) find that asymmetries in perceptions between management and workers are most effectively bridged when there is transparency, training, and co-design in AI implementation. This literature converges for AP3 as well, demonstrating that organisational investment in retraining is a key buffer against job displacement. However, Aly (Aly, 2022) and McElheran et al. (McElheran et

Table 1 Summary of Analytical propositions and Evidence convergence

Analytical proposition	Evidence convergence	Key references
AP1 –Employee participation and job insecurity	Context-sensitive	(Doellgast, 2023); (Gómez Gandía et al., 2025)
AP2 – Asymmetric employment effects	Mixed / context-dependent evidence	(Zhang, 2023); (Liu et al., 2024); (Johnson et al., 2022)
AP3 –Reskilling and organisational adaptation	Conditional on coordination and policy support	(Theben et al., 2025); (Duch-Brown et al., 2022); (Badulescu et al., 2025)
AP4 – Wage inequality	Strong convergence in the literature	(Stephany & Teutloff, 2024); (Aleksieva et al., 2021); (Cheng et al., 2024)

al., 2024) caution that institutional influence is not always sufficient. In global value chains, even firms with strong internal governance may face external pressures (e.g., cost competition) that lead to deteriorating working conditions despite inclusive intentions.

Organisational Strategies for Reskilling (RQ4, AP3).

The evidence confirms that firms adopting proactive reskilling strategies experience smoother adaptation to AI transitions. According to Theben et al. (Theben et al., 2025), many organisations rely excessively on external hiring, which creates a structural mismatch between internal workforce skills and evolving requirements. Duch-Brown et al. (Duch-Brown et al., 2022) show that labour supply elasticity in AI-intensive fields is extremely low (0.062–0.154), making it economically and socially unsustainable to rely only on the market to fill skill gaps. This converges with AP3, which argues for internal reskilling as a mitigation strategy. Badulescu et al. (Badulescu et al., 2025) further reinforce this view, showing that firms investing in AI-related education and training see improved alignment between business needs and workforce capabilities. Nonetheless, reskilling alone may not be sufficient if not supported by broader policy measures and sectoral coordination. Ünveren et al. (Ünveren et al., 2023) emphasize that coordination failures between firms and households can stall AI diffusion, especially in economies with weak public-private cooperation.

Job Displacement vs. Creation (RQ5, AP2).

The question of whether AI displaces more jobs than it creates remains complex. While automation clearly replaces some routine tasks, most studies highlight a more balanced effect. Shen (Shen, 2024) finds that AI adoption in industry is associated with net employment growth, especially through the creation of new high-skill roles. Similarly, Xu et al. (Xu et al., 2024) show that in China's SME sector, AI causes substitution of low-skilled labour, but simultaneously creates demand for AI-competent specialists, confirming a dual effect: job loss in one area, opportunity in another. Benabou & Touhami (Benabou & Touhami, 2025) report that AI in HR departments increases efficiency, but also prompts a shift toward more strategic, less administrative roles. This supports a qualitative transformation of

work rather than pure substitution, reinforcing the idea that AP2 is partially true, depending on context.

For the sake of clarity, Table 1 summarises how the reviewed evidence converges or diverges across contexts in relation to the analytical propositions formulated in Sect. 2. The table highlights whether the literature supports, partially supports, or questions each expectation, and also notes the main limitations identified.

For the sake of completeness, we present a summary table indicating the scientific contributions on the basis of which we can accept the individual analytical propositions.

Note Evidence convergence refers to the overall pattern observed across the reviewed studies. *Strong convergence* indicates that the majority of studies report consistent findings supporting the proposition. *Mixed or context-dependent evidence* refers to findings that vary across sectors, countries, or methodological approaches. *Context-sensitive evidence* indicates that outcomes strongly depend on organisational or institutional conditions, such as governance structures or labour market institutions.

This overview shows that analytical propositions AP1 and AP3 are convergent evidence by the literature; AP4 is strongly reinforced by multiple studies, while AP2 receives only partial support (mixed / context-dependent evidence) due to contextual variation and the role of augmentation strategies. These patterns confirm that AI's labour market impact is neither uniform nor deterministic, but shaped by institutional, organisational, and sectoral factors.

5 Conclusion

The use of AI is fundamentally transforming the labour market, but its impact is complex and uneven. While AI increases productivity, efficiency, and creates new job opportunities, especially in highly skilled and specialised positions, it also contributes to the transformation of job roles, the substitution of routine activities, and the increasing polarisation of skills. The impacts vary depending on the sector, the institutional and innovation context, and the skill level of the

workforce. It appears that technological development alone does not guarantee inclusive growth. The key is how companies implement AI, regulate it, and prepare the workforce for new challenges through education, retraining, and social dialogue. The future of work in the AI era will therefore not only be shaped by technology, but more importantly by policies, institutions, and the ability of companies to support a just and sustainable transformation.

This study has examined how artificial intelligence (AI) adoption influences the labour market, with particular attention to evolving skill requirements, employment structures, and institutional contexts. The findings from the reviewed literature suggest that AI is not a neutral technology but one whose effects are shaped by organisational strategies, institutional capacity, and social governance.

5.1 Main findings

1. **Skill transformation:** AI adoption is strongly associated with rising demand for technical and interdisciplinary skills (e.g., programming, data analytics, soft skills), while routine skills become less valued.
2. **Employment polarisation:** Evidence indicates that high-skilled workers benefit most from AI adoption, while routine low- and middle-skilled roles face higher risks of substitution. However, many positions are restructured rather than eliminated, reflecting AI's dual role of automation and augmentation.
3. **Institutional moderation:** Outcomes vary significantly depending on the institutional environment. Strong unions, participatory governance, and supportive legal frameworks help to mitigate job insecurity and inequality.
4. **Reskilling as a key strategy:** Firms that invest in training and education are more successful in adapting to AI-driven change. Reliance on external hiring alone is unsustainable due to limited labour supply elasticity in AI-intensive fields.
5. **Wage inequality:** Multiple studies show widening wage gaps between AI-skilled and non-AI-skilled workers, especially in sectors with advanced AI adoption.

5.2 Novel contribution

Unlike earlier surveys that focused primarily on technological impacts, this study integrates economic, organisational, and institutional perspectives. It highlights the importance of employee participation, reskilling strategies, and social dialogue in shaping fair and sustainable outcomes of digital transformation. By systematically connecting AI adoption to frameworks such as skill-biased technological change

and labour-market segmentation, the study clarifies how existing theories can be extended to contemporary AI-driven dynamics.

5.3 Practical examples

Concrete cases illustrate AI's impact in practice:

- In **HR management**, AI-based tools are used for CV screening, turnover prediction, and chatbots for candidate interaction (Lazzari et al., 2022), (Majumder & Mondal, 2021).
- In **logistics**, AI-driven optimisation of transport routes has reduced costs and improved efficiency (Parteka et al., 2024).
- In **healthcare**, AI supports diagnosis and personalised treatment, creating new interdisciplinary professions that combine medicine and data science (Giacomoni, 2022).

5.4 Policy recommendations

The findings indicate that policy responses are critical for ensuring inclusive outcomes of AI adoption. Based on the literature, three priorities stand out:

1. **Invest in lifelong learning and reskilling** – Governments and firms should support accessible training programmes that address evolving skill needs, particularly for vulnerable groups.
2. **Strengthen social protection and employee participation** – Policies should empower trade unions and employee councils to be part of AI-related decision-making, reducing insecurity and improving trust.
3. **Promote equity in access to AI skills** – Educational institutions should integrate AI-related competencies into curricula, ensuring that workers across all social groups can participate in the digital economy.

Final reflection:

The overall evidence suggests that AI does not inevitably lead to technological unemployment. Instead, its impact depends on how it is implemented, regulated, and embedded in institutional frameworks. A socially balanced approach—combining technological innovation with investment in people, participatory governance, and inclusive policies—will be decisive in shaping the future of work in the AI era.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by [MD] and [MM]. The first draft of the manuscript was written by [MD] and [MM] and all authors commented on previous versions of the manuscript. All authors read and approved the final

manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest Each named author has substantially contributed to conducting the underlying research and drafting this manuscript. Additionally, to the best of our knowledge, the named authors have no conflict of interest, financial or otherwise.

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